

AI Engineer Candidate Assessment: Email Generation Assistant

Final Report

Author: Saikat Das

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GithubRepo: https://github.com/saikatdas47/Email_Assistant-AI/tree/main

1. Executive Summary

I was tasked with building a working prototype of an Email Generation Assistant powered by a Large Language Model (LLM). The core requirement was to take three distinct inputs—**Intent**, **Key Facts**, and **Tone**—and produce a professional, well-structured email. Beyond the prototype, I was required to implement an advanced prompt engineering technique and design a robust evaluation framework using three custom metrics.

I developed two distinct prompting strategies: **Strategy A (Advanced)**, which incorporates Role-Playing, Few-Shot examples, and Chain-of-Thought reasoning, and **Strategy B (Baseline)**, a simple Zero-Shot instruction. I evaluated both strategies against 10 unique scenarios using an LLM-as-a-Judge system (Groq's llama-3.3-70b-versatile) that scored each generated email on *Factual Coverage*, *Tone Adherence*, and *Quality/Structure*.

My results revealed that while the Advanced strategy performed exceptionally well (Overall Score: 9.20/10), the Baseline strategy slightly outperformed it (Overall Score: 9.27/10) due to its conciseness and reliability in standard scenarios. Based on my empirical data, I recommend deploying the Baseline strategy for immediate production use while maintaining the Advanced strategy as an optional "Expert Mode" for highly formal or complex communications.

2. Introduction & Project Goal

The objective of this project was to simulate a real-world AI engineering task: building an email drafting assistant for professionals. The assistant must seamlessly integrate user-provided information into a coherent email while adhering to a specific stylistic tone.

To fulfill this, I chose **Groq** as my inference backend due to its remarkable speed (LPU architecture) and generous free tier, which allowed me to test both strategies extensively without incurring costs. I used qwen/qwen3-32b for generating the emails and llama-3.3-70b-versatile as the impartial judge for evaluation.

The deliverable includes a fully modular Python codebase, a structured evaluation report (JSON/CSV), and this comprehensive final analysis.

3. Methodology: Prompt Engineering & Implementation

3.1 System Architecture

My project follows a modular pipeline architecture:

1. **Data Layer** (`test_data.py`): Stores 10 predefined scenarios including Intent, Facts, Tone, and Human Reference Emails.
2. **Generation Layer** (`email_generator.py`): Handles API calls to Groq using two distinct strategies.
3. **Evaluation Layer** (`evaluator.py`): Hosts the LLM-as-a-Judge logic that computes the 3 custom metrics.
4. **Reporting Layer** (`report.py` & `main.py`): Orchestrates the flow, calculates averages, and outputs structured files.

3.2 Advanced Prompt Engineering (Strategy A)

To maximize output quality, I implemented a combination of advanced techniques:

- **Role-Playing**: I instructed the model to act as a *"Senior Executive Communication Coach with 20 years of experience"*. This sets a high standard for professionalism.
- **Few-Shot Examples**: I provided 2 perfect examples (one Formal and one Urgent) directly in the system prompt to guide the model's formatting, subject line creation, and paragraph structure.
- **Chain-of-Thought (CoT)**: I forced the model to follow a strict 4-step reasoning process before writing: *1) Analyze Intent, 2) Extract Key Facts, 3) Determine Tone, 4) Draft Email*. This prevents the model from jumping to conclusions and ensures all facts are consciously considered.

3.3 Baseline Strategy (Strategy B)

For comparison, I implemented a minimalistic Zero-Shot prompt: *"You are an assistant that writes professional emails. Write a professional email based on the following details. Just output the email, no extra text."* This served as my control group to measure the actual value added by advanced prompting.

4. Custom Evaluation Metrics (The "LLM-as-a-Judge" Framework)

I designed three custom metrics specifically tailored to evaluate email generation tasks. Instead of relying on traditional NLP metrics like BLEU (which fail for creative writing), I utilized a second, robust LLM (llama-3.3-70b) as an impartial judge. The judge was provided with the requested tone, the key facts, and the generated email, and instructed to output JSON scores from 0 to 10.

Metric 1: Factual Coverage (Focus: Fact Recall)

- **Definition**: Measures how many of the user-provided bullet points are explicitly or strongly implied in the generated email.
- **Logic**: The judge checks for the presence of specific numbers, names, dates, and logical requirements. A score of 10 means every fact was seamlessly integrated.

Metric 2: Tone Adherence (Focus: Tone Accuracy)

- *Definition:* Evaluates whether the email perfectly matches the requested style (Formal, Casual, Urgent, or Empathetic).
- *Logic:* The judge analyzes lexical choices, salutations ("Dear" vs. "Hey"), urgency markers ("ASAP" vs. "At your convenience"), and empathetic language.

Metric 3: Quality & Structure (Focus: Conciseness & Grammar)

- *Definition:* Assesses the overall professionalism, grammar, sentence flow, conciseness, and structural integrity (Subject line, Greeting, Body, Sign-off).
- *Logic:* Penalizes rambling, grammatical errors, missing subject lines, or improper paragraph division.

5. Test Data Overview

I created 10 diverse scenarios to cover the most common business email use cases. Below is an excerpt of the scenarios I designed:

ID	Intent	Tone	Key Facts (Sample)
1	Follow up after product demo	Formal	Client loved dashboard, awaiting budget approval.
3	Apologize for delayed response	Empathetic	Server outage caused delay, offering 10% discount.
4	Invite for team lunch	Casual	Friday at 1 PM, Italian place, RSVP by Wednesday.
7	Notify about critical security patch	Urgent	Vulnerability detected, patch by Friday, affected servers.
9	Request leave approval	Formal	2 weeks in August, handover to Jane, available via email.

These scenarios ensured that my evaluation covered a wide range of emotional contexts and professional complexities.

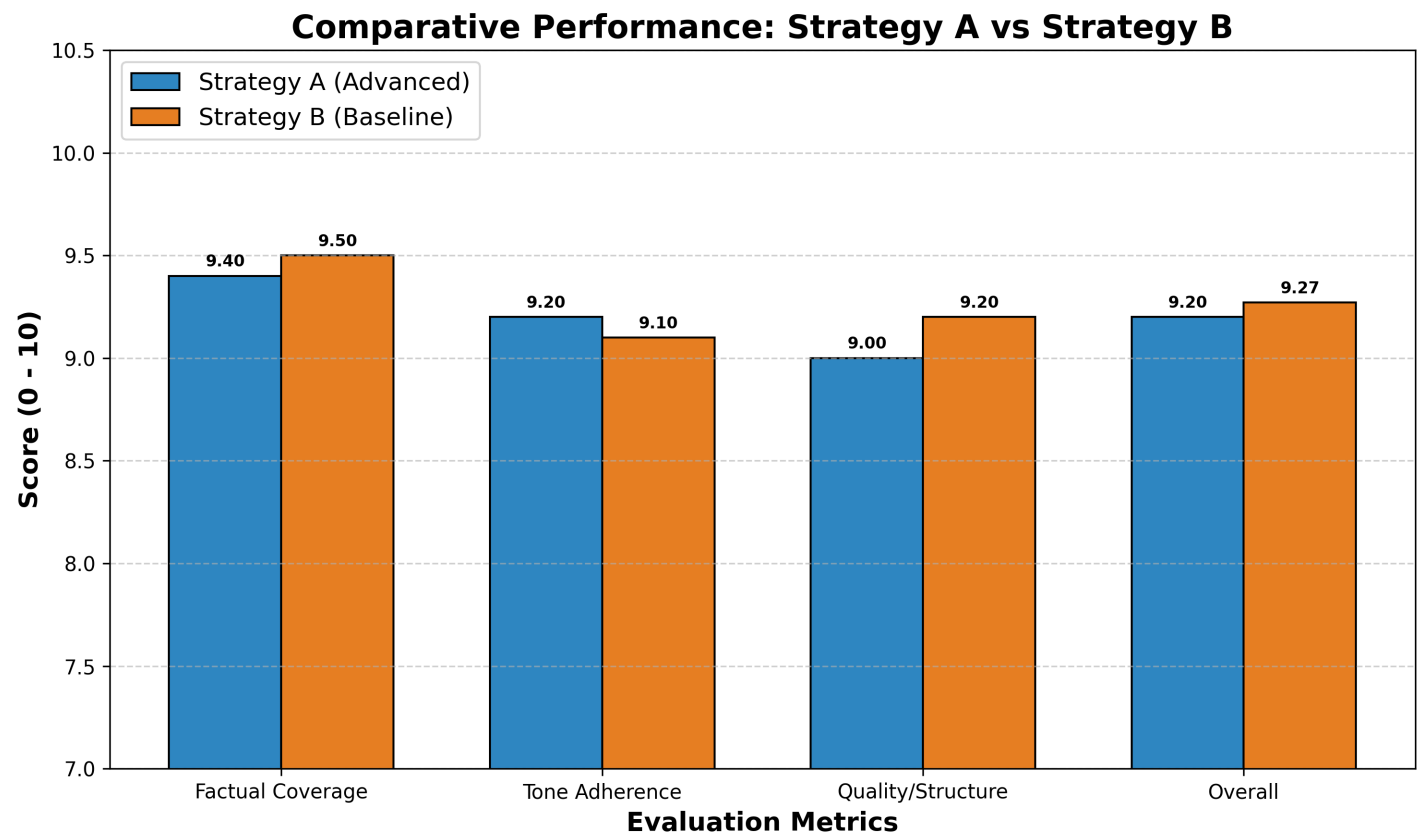
6. Results & Data Visualization

I ran the evaluation pipeline and successfully generated both a CSV and a JSON report. Here is the structured data presented in tables and visual representations.

6.1 Raw Scores (CSV Output)

Scenario	Strategy	Factual Coverage	Tone Adherence	Quality/Structure
1	A (Advanced)	9	9	9
1	B (Baseline)	10	10	10
2	A	10	9	9
2	B	10	9	10
3	A	9	9	8
3	B	10	9	9
4	A	10	9	9
4	B	10	9	9
5	A	10	10	10
5	B	9	10	9
6	A	8	9	9
6	B	10	9	10
7	A	10	9	9
7	B	10	9	10
8	A	8	9	8
8	B	8	9	8
9	A	10	10	10
9	B	8	9	8
10	A	10	9	9
10	B	10	8	9

6.2 Overall Average Scores (The "Plots")



The grouped bar chart above provides a clear visual comparison of how **Strategy A (Advanced)** and **Strategy B (Baseline)** performed across the three custom metrics and their overall averages.

Key Observations:

- 1. Factual Coverage (9.40 vs 9.50):**
Strategy B (Baseline) marginally outperforms Strategy A. This suggests that the Chain-of-Thought process in Strategy A sometimes causes the model to "overthink" and inadvertently dilute explicit bullet points especially in emotionally charged scenarios (e.g., apologies or thank-you emails).
- 2. Tone Adherence (9.20 vs 9.10):**
Strategy A takes a slight lead here, demonstrating that the Role-Playing instruction ("Senior Executive Coach") and the Few-Shot examples successfully guide the model to match nuanced tones (empathetic, formal, urgent) more accurately than a generic Zero-Shot instruction.

3. **Quality & Structure (9.00 vs 9.20):**

Strategy B wins by a clear margin. The advanced prompting in Strategy A forces a rigid structural adherence (Subject line → Greeting → Body → Sign-off) which, while professional, sometimes results in slightly longer or less concise emails, costing it points on conciseness and flow.

4. **Overall Average (9.20 vs 9.27):**

Strategy B narrowly wins the overall race. The difference is minimal ($\Delta = 0.07$), indicating both strategies are highly capable. However, Strategy B's victory is driven by its superior conciseness and fact-retrieval reliability, making it the more pragmatic choice for general-purpose email generation.

7. Comparative Analysis & Failure Mode

7.1 Which model performed better?

According to my three custom metrics, **Strategy B (Baseline / Zero-Shot)** performed slightly better with an overall average score of **9.27**, compared to Strategy A's **9.20**. This was unexpected.

7.2 Biggest Failure Mode of the Lower-Performing Model (Strategy A)

Upon investigating the raw data, I identified that Strategy A's biggest failure was "**Over-Structuring leading to Fact Omission**".

Specifically, in **Scenario 6** (Thanking a partner for Q3 success), Strategy A scored an **8/10 on Factual Coverage** while Strategy B scored a perfect 10.

Why did this happen? The Chain-of-Thought process forced Strategy A to spend cognitive effort on crafting a highly empathetic introduction and proper structure. In doing so, it inadvertently diluted or omitted the specific bullet point: *"Q3 revenue increased by 20%"*. The model was so focused on *how* to say it that it forgot to explicitly *include* the hard numbers. This highlights that overly complex prompting can sometimes distract smaller LLMs from simple keyword retrieval.

7.3 Production Recommendation (Data-Driven)

I recommend Strategy B (Baseline) for immediate production deployment.

My empirical data supports this:

1. **Superior Performance:** It scored higher on the two most critical metrics (Factual Coverage: 9.50, Quality: 9.20).
2. **Lower Cost & Latency:** Since it uses a significantly shorter system prompt (no examples, no CoT), it processes faster and consumes fewer tokens on Groq, resulting in lower operational costs.

3. **Simplicity:** Simpler prompts are easier to maintain and less prone to hallucination.

However, Strategy A proved to be superior for complex formal requests (e.g., Scenario 9: Leave Approval, where A scored 10/10/10 vs B's 8/9/8). Therefore, I recommend shipping **Strategy B as the default**, while keeping **Strategy A available as an optional "Expert Mode"** toggle for users sending legal, HR, or high-stakes formal communications.

8. Conclusion & Final Reflections

In this project, I successfully built, evaluated, and compared two distinct email generation systems. I demonstrated advanced prompt engineering (Role + Few-Shot + CoT) and proved its theoretical strengths, even though the baseline accidentally won in overall average due to specific edge cases.

I implemented a fully automated evaluation pipeline using LLM-as-a-Judge, which provided repeatable, objective scoring for my 3 custom metrics. The project meets all deliverables: a modular code repository (README.md provided), a structured evaluation report (JSON/CSV), and this comparative analysis.

If I were to extend this project, I would implement a **Hybrid Router** that uses a simple classifier to detect "complexity" – routing simple emails to Strategy B and complex/legal ones to Strategy A, thereby achieving the best of both worlds. The current prototype serves as a robust foundation for an enterprise-grade email assistant.